Sequence

A function: $N \to \mathbb{R}$, $n \mapsto f(n) = x_n$.

Denoted: $\{x_n\}_{n\in N}$, $\{x_n\}_{n=1}^{\infty}$, $\{x_n\}$

Def 2.1: Convergence:

A sequence $\{x_n\}$ converges to $a \in \mathbb{R}$ if for any $\varepsilon > 0$ there exists $N \in N$ such that

$$n \geq N \Rightarrow |x_n - a| < \varepsilon$$

"$\forall \varepsilon > 0,\ \exists N \in \mathbb{N}\ \ n \geq N \Rightarrow |a_n - a| < \varepsilon$"

We write: $x_n \to a$ or $\lim_{n\to\infty} x_n = a$.

$x_n \to a \Leftrightarrow x_n - a \to 0 \Leftrightarrow |x_n - a| \to 0$.

e.x. proving $1/n \to 0$.

$\forall \varepsilon > 0$ $\exists N \in \mathbb{N}$. $N > \frac{1}{\varepsilon}$.

$\forall n \geq N$ $\overset{|a_n - a|}{|\frac{1}{n} - 0|} = \frac{1}{n} \leq \frac{1}{N} < \varepsilon$, proved.

e.x. Suppose $x_n \to 2$. . prove that $\frac{2x_n+1}{x_n} \to \frac{5}{2}$.

$\varepsilon > 0$. need N. such that $\forall \varepsilon > 0$ $\quad n \geq N \Rightarrow |\frac{2x_n+1}{x_n} - \frac{5}{2}| < \varepsilon$

$|\frac{2x_n+1}{x_n} - \frac{5}{2}| > |\frac{1}{x_n} - \frac{5}{2}| = |\frac{1}{x_n} - \frac{1}{2}| = \frac{|2-x_n|}{2|x_n|} <$ () $< \varepsilon$ ← idea: find this

(1) Since $x_n \to 2$. for $\varepsilon_1 = 1$ $\exists N_0 > 0$ such that $n \geq N_0 \Rightarrow |x_n - 2| < 1 \Rightarrow x_n \in (1,3)$

$\Rightarrow x_n > 1$

(2) Since $x_n \to 2$. for $\forall 2\varepsilon > 0$ $\quad \exists N_1 > 0$. $n \geq N_1 \Rightarrow |x_n - 2| \leq 2\varepsilon$

So if $n \geq N$: $\max\{N_0, N_1\}$ then $x_n > 1$ and $|x_n - 2| < 2\varepsilon$.

Such that $|\frac{2x_n+1}{x_n} - \frac{5}{2}| = \frac{|2-x_n|}{2|x_n|} < \frac{|x_n-2|}{2|x_n|} < \frac{2\varepsilon}{2} = \varepsilon$.

A convergent sequence has only 1 limit

if $x_n \to a$ and $x_n \to b$ then $a = b$

proof: if $a \neq b$. Assume $a > b$.

$\varepsilon = a - b > 0$.

$\exists N_1$ $\quad n \geq N_1 \Rightarrow |x_n - a| < \frac{\varepsilon}{2}$

$\exists N_2$ $\quad n \geq N_2 \Rightarrow |x_n - b| < \frac{\varepsilon}{2}$

if $n = \max\{N_1, N_2, b\text{ both above hold}\}$.

triangle inequality: $\varepsilon = |a-b| = |x_n - a + b - x_n| < |x_n - a| + |x_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$

$\Rightarrow \varepsilon < \varepsilon$. contradiction

Negating the definition

$x_n \to a$: $\forall \varepsilon > 0$ $\exists N \in \mathbb{N}$ $\forall n \geq N$: $|x_n - a| < \varepsilon$.

$x_n \not\to a$ $\exists \varepsilon_0 > 0$ $\forall N \in \mathbb{N}$ $\exists n_0 \geq N$: $|x_n - a| \geq \varepsilon_0$.

e.x. prove $(-1)^n$ does not converge

proof: suppose $(-1)^n$ converges to a for some a.

for $\varepsilon = 1$ $\exists N > 0$. $\forall n > N$ $|(-1)^n - a| < 1$.

But take $n > 2N$ which is even.

$|(-1)^n - a| = |1 - a| < 1 \Rightarrow 0 < a < 2$

take $n > 2N+1$ which is odd.

$|(-1)^n - a| + |-1 - a| > |N+1| < 1 \Rightarrow -2 < a < 0$

Since these two interval do not intersect: contradiction.

For Subsequence

remember: $n_k \geq k$

if $x_n \to a$. then every subsequence $\{x_{n_k}\}$ also converges to a

proof: $\forall \varepsilon > 0$ $\exists N > 0$ $\quad n \geq N \Rightarrow |x_n - a| < \varepsilon$.

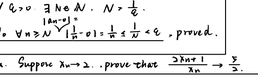
Consider subsequence $\{x_{n_k}\}$

Since $n_k \geq k$ so if $k \geq N$ then $n_k \geq k \geq N$.

$k \geq N \Rightarrow n_k \geq N \Rightarrow |x_{n_k} - a| < \varepsilon$

if every subsequences converge to a then the original Sequence converge to a.

Bounded Sequence



Thm 2.1 Every Convergent Sequence is bounded

Suppose $x_n \to a$ as $n \to \infty$

Take $\varepsilon = 1$. $\exists N > 0$. $\quad n \geq N \Rightarrow |x_n - a| < 1$.

$|x_n| = |x_n - a + a| \leq |x_n - a| + |a| < 1 + |a|$.

$\Rightarrow |x_n| \leq \max\{1 + |a|, |a_1|, |a_2|, \cdots |a_n|\}$.

Bounded Sequence may not converge. (e.g. $a_n = (-1)^n$)

Limit Theorem

Squeeze Theorem

(1) if $x_n \to a$ $y_n \to a$. and there is $N \in \mathbb{N}$ that $n \geq N_0 \Rightarrow x_n \leq w_n \leq y_n$ then $w_n \to a$.

(2) if $x_n \to 0$ and $\{y_n\}$ is bounded, then $x_n y_n \to 0$

proof: Since $x_n \to a$ $y_n \to a$. $\forall \varepsilon > 0$ $\exists N_1, N_2 > 0$.

$n \geq N_1 \Rightarrow |x_n - a| < \varepsilon \Rightarrow a - \varepsilon < x_n$ (1)

$n \geq N_2 \Rightarrow |y_n - a| < \varepsilon \Rightarrow y_n < a + \varepsilon$ (2)

Thus: if $n \geq N_3 = N_1 + N_2 + N_0$ then (1). (2) hold.

$a - \varepsilon < x_n \leq w_n \leq y_n < a + \varepsilon$. and: $x_n \leq w_n \leq y_n$

$\Rightarrow |w_n - a| < \varepsilon$

Hence w_n converge to a.

proof: let $|y_n| < M$ $\forall n \in \mathbb{N}$ for some $m > 0$.

Then $\forall \varepsilon > 0$ since $x_n \to 0$ $\exists N > 0$

$n \geq N \Rightarrow |x_n| < \frac{\varepsilon}{M}$

So $n \geq N \Rightarrow |x_n y_n| \leq |x_n \cdot M| = M|x_n| < M \cdot \frac{\varepsilon}{|M|} = \varepsilon$.

e.x. find $\lim_{n\to\infty} 2^{-n} \cos(n^3 \cdot n^2 + n - 1)$

(1) $2^{-n} \to 0$ as $n \to \infty$.

(2) $\cos(n^3 \cdot n^2 + n - 1)$ is bounded $\Rightarrow \lim_{n\to\infty} 2^{-n} \cos(n^3 \cdot n^2 + n - 1) = 0$

$-1 \leq \cos(n^3 \cdot n^2 + n - 1) \leq 1$

proof: $\forall \varepsilon > 0$. $\exists N \in \mathbb{N}$, $N > \log \frac{1}{\varepsilon}$

$n \geq N \Rightarrow |a_n - a| = |\frac{1}{2^n}| = \frac{1}{2^n} < \frac{1}{2^N} < \frac{1}{2^{\log \frac{1}{\varepsilon}}} < \varepsilon$

$\therefore a_n \to 0$ by definition

Theorem 2.11: approximating a supremum or infimum

Let $\varnothing \neq E \subseteq \mathbb{R}$.
(i) If E has a finite supremum, there is a sequence $\{x_n\} \subseteq E$ such that $x_n \to \sup E$.
(ii) If E has a finite infimum, there is a sequence $\{y_n\} \subseteq E$ such that $y_n \to \inf E$.

proof: (i) Since E has a finite supremum

$\exists x_n \in E$. $S - \frac{1}{n} < x_n \leq S$ for any $n \in \mathbb{N}$

Since $S - \frac{1}{n} \to S$ as $n \to \infty$.

by squeeze theorem. $x_n \to S$ as $n \to \infty$.

(ii) $\exists x_n \in E$. $S \leq x_n < S + \frac{1}{n} \Rightarrow S + \frac{1}{n} \to S \Rightarrow$ Squeeze theorem $\Rightarrow x_n \to S$

Theorem 2.12: algebra of finite limits

Suppose that $x_n \to a$ and $y_n \to b$, where $a, b \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. Then
(i) $x_n + y_n \to a + b$;
(ii) $\alpha x_n \to \alpha a$;
(iii) $x_n y_n \to ab$;
(iv) if $y_n \neq 0$ for every n and $b \neq 0$, then $\frac{x_n}{y_n} \longrightarrow \frac{a}{b}$.

(i) $\forall \varepsilon > 0$. $\exists N_1 \in \mathbb{N}$. $\quad n \geq N_1$ $\quad |x_n - a| < \frac{\varepsilon}{2}$

$\exists N_2 \in \mathbb{N}$ $\quad n \geq N_2$ $\quad |y_n - b| < \frac{\varepsilon}{2}$

$\Rightarrow$ (let $N_3 = \max\{N_1, N_2\}$)

$n \geq N_3$. $\Rightarrow |x_n + y_n - (a+b)| = |x_n - a + (y_n - b)| \leq |x_n - a| + |y_n - b| < \varepsilon$.

$\Rightarrow x_n + y_n \to a + b$ as $n \to \infty$

(ii) $\forall \varepsilon > 0$. $\exists N \in \mathbb{N}$. $\quad n \geq N$ $\quad |x_n - a| < \frac{\varepsilon}{|\alpha|}$

$\Rightarrow |\alpha x_n - a| = |\alpha||x_n - a| < |\alpha| \cdot \frac{\varepsilon}{|\alpha|} = \varepsilon$

$\Rightarrow \alpha x_n \to \alpha a$ as $n \to \infty$

(iii) since $x_n \to a$. $y_n \to b$. by theorem 2.8 x_n, y_n is bounded

$\Rightarrow |x_n| \leq M$. $|b| \leq M$ for some M

$\forall \varepsilon > 0$. $\exists N_1 \in \mathbb{N}$. $\quad n \geq N_1$ $\quad |x_n - a| < \frac{\varepsilon}{2M}$

$\exists N_2 \in \mathbb{N}$ $\quad n \geq N_2$ $\quad |y_n - b| < \frac{\varepsilon}{2M}$

$|x_n y_n - ab| = |x_n(y_n - b) + b(x_n - a)|$

$\leq |x_n||y_n - b| + |b||x_n - a|$

$\leq M \cdot \frac{\varepsilon}{2M} + M \cdot \frac{\varepsilon}{2M} = \varepsilon$. $\Rightarrow x_n y_n \to ab$

(iv) ★ first. we prove $\frac{1}{y_n} \to \frac{1}{b}$ as $n \to \infty$.

① $\exists N_1 \in \mathbb{N}$. $\quad n \geq N_1 \Rightarrow |y_n - b| < \frac{b}{2} \Rightarrow y_n > \frac{b}{2}$

② $\exists N_2 \in \mathbb{N}$ $\quad n \geq N_2 \Rightarrow |y_n - b| < \frac{b^2 \cdot \varepsilon}{2}$

$n \geq \max\{N_1, N_2\} \Rightarrow |\frac{1}{y_n} - \frac{1}{b}| = \frac{|y_n - b|}{|b||y_n|} < \frac{\frac{\varepsilon b^2}{2}}{b \cdot \frac{b}{2}} = \varepsilon$.

$\Rightarrow \frac{1}{y_n} \to \frac{1}{b}$ as $n \to \infty$.

By (iii), $\frac{x_n}{y_n} = x_n \cdot \frac{1}{y_n} \to \frac{a}{b}$

Prove: $\lim_{n\to\infty} \frac{n^3 + n^2 - 1}{1 - 3n^3} = -\frac{1}{3}$ By thm 2.12.

$\lim_{n\to\infty} \frac{n^3 + n^2 - 1}{1 - 3n^3} = \lim_{n\to\infty} \frac{1 + \frac{1}{n} - \frac{1}{n^3}}{\frac{1}{n^3} - 3}$

By definition of limit. we can prove $\frac{1}{n} \to 0$. $\frac{1}{n^3} \to 0$ as $n \to \infty$.

$\Rightarrow \lim_{n\to\infty} \frac{1 + \frac{1}{n} - \frac{1}{n^3}}{\frac{1}{n^3} - 3} = \lim_{n\to\infty} \frac{1 + 0 - 0}{0 - 3} = -\frac{1}{3}$.

Definition 2.14: infinite limits

Let $\{x_n\}$ be a real sequence.
(i) We write $x_n \to +\infty$ if, for every $M \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that
$$n \geq N \implies x_n > M.$$
(ii) We write $x_n \to -\infty$ if, for every $M \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that
$$n \geq N \implies x_n < M.$$
In either case the sequence diverges in $\mathbb{R}$, although it has a limit in the extended real line.

Unbounded sets contain divergent sequences

Let $E \subseteq \mathbb{R}$.
(i) If $\sup E = +\infty$, there is a sequence $\{x_n\} \subseteq E$ such that $x_n \to +\infty$.
(ii) If $\inf E = -\infty$, there is a sequence $\{y_n\} \subseteq E$ such that $y_n \to -\infty$.

The extended real number

Define: $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$

For $\phi \neq E \subseteq \mathbb{R}$: $\sup E = +\infty$ if E is unbounded above; $\inf E = -\infty$ if E is unbounded below

For $E = \phi$: $\sup E = -\infty$. $\inf E = \infty$.

proof: ① if $\sup E = \infty$, E is unbounded above

For every $n \in \mathbb{N}$, choose $x_n \in E$. with $x_n > n$. → Construct a Sequence

Given $M \in \mathbb{R}$ choose $N > M$. For $n \geq N$

$x_n > n \geq N > M$

By def 2.14. $x_n \to \infty$ as $n \to \infty$.

② if $\inf E = -\infty$. Choose $x_n < -n$

$x_n < -n \leq -N < -M \Rightarrow x_n \to -\infty$ as $n \to \infty$

Theorem 2.15: arithmetic with infinite limits

Suppose that $x_n \to +\infty$; the corresponding conclusions for $x_n \to -\infty$ are shown in parentheses.
(i) If $\{y_n\}$ is bounded below (respectively, bounded above), then
$$x_n + y_n \to +\infty \qquad \text{(respectively, } x_n + y_n \to -\infty\text{).}$$
otherwise may $y_n \to -\infty$
(ii) If $\alpha > 0$, then
$$\alpha x_n \to +\infty \qquad \text{(respectively, } \alpha x_n \to -\infty\text{).}$$
(iii) If $y_n \geq M_0$ for some $M_0 > 0$ and every n, then
$$x_n y_n \to +\infty \qquad \text{(respectively, } x_n y_n \to -\infty\text{).}$$
(iv) If $\{y_n\}$ is bounded and $x_n \neq 0$ for every n, then
$$\frac{y_n}{x_n} \longrightarrow 0.$$

proof: (i) Suppose $y_n \geq L$ for all n Given $K \in \mathbb{R}$

$x_n \to +\infty \Rightarrow x_n > k - L$

$x_n + y_n > L + k - L = k \Rightarrow x_n + y_n \to +\infty$

(ii) $x_n \to +\infty \Rightarrow x_n > \frac{k}{\alpha}$ for any given $k \in \mathbb{R}$.

$\alpha x_n > \alpha \cdot \frac{k}{\alpha} = k \Rightarrow \alpha x_n \to +\infty$

(iii) $x_n \to \infty$. $\Rightarrow x_n > \max\{0, \frac{k}{M_0}\}$.

$x_n y_n > \frac{k}{M_0} \cdot M_0 = k$. $\Rightarrow x_n y_n \to +\infty$

(iv) y_n is bounded $\Rightarrow |y_n| \leq B$.

$x_n \to +\infty$. $x_n > \frac{B}{\varepsilon}$ for any given $\varepsilon \in \mathbb{R}$

$|\frac{y_n}{x_n} - 0| = \frac{|y_n|}{|x_n|} \leq \frac{B}{|x_n|} < \varepsilon \Rightarrow \frac{y_n}{x_n} \to 0$.

The extended real line

Write
$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$
The following conventions are consistent with Theorem 2.15:

Addition	Multiplication
$x + (+\infty) = +\infty$	$x(+\infty) = +\infty$ if $x > 0$
$x + (-\infty) = -\infty$	$x(-\infty) = -\infty$ if $x > 0$
$(+\infty) + (+\infty) = +\infty$	$x(+\infty) = -\infty$ if $x < 0$
$(-\infty) + (-\infty) = -\infty$	$x(-\infty) = +\infty$ if $x < 0$
	$(+\infty)(+\infty) = (-\infty)(-\infty) = +\infty$
	$(+\infty)(-\infty) = (-\infty)(+\infty) = -\infty$

Indeterminate forms

Expressions such as
$$(+\infty) - (+\infty), \qquad \frac{\infty}{\infty}, \qquad 0 \cdot \infty$$
are not assigned values. Different sequences producing the same symbolic form can have different limits, so each case requires further analysis.

Corollary 2.16: extended-real limit arithmetic

Let $x_n \to x$ and $y_n \to y$, where $x, y \in \overline{\mathbb{R}}$.
(i) $x_n + y_n \to x + y$, provided $x + y$ is not of the form $(+\infty) + (-\infty)$.
(ii) For $\alpha \in \mathbb{R}$, $\alpha x_n \to \alpha x$, provided the product is not of the form $0 \cdot (\pm\infty)$.
(iii) $x_n y_n \to xy$, provided the product is not of the form $0 \cdot (\pm\infty)$.

Theorem 2.17: comparison theorem

Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences. If there is $N_0 \in \mathbb{N}$ such that
$$x_n \leq y_n \qquad (n \geq N_0),$$
then
$$\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n.$$
In particular, if $x_n \in [r, s]$ for every n and $x_n \to c$, then $c \in [r, s]$.

proof: Suppose $x_n \to a$. $y_n \to b$. and $a > b$ by contradiction

let $\delta = a - b > 0$.

$\exists N_1 \cdot N_2 \in \mathbb{N}$. $\quad n \geq N_1 \Rightarrow |x_n - a| < \frac{\delta}{2}$

$\quad n \geq N_2 \Rightarrow |y_n - b| < \frac{\delta}{2}$.

→ for $n \geq \max\{N_0, N_1, N_2\}$

$\delta = a - b = (a - x_n) + (x_n - y_n) + (y_n - b)$

$< \frac{\varepsilon}{2} + 0 + \frac{\varepsilon}{2} = \varepsilon$.

Contradiction. hence $a \leq b$.

Strict inequalities may disappear in the limit

Even if $x_n < y_n$ for every sufficiently large n, one can conclude only that
$$\lim x_n \leq \lim y_n,$$
not that the inequality between the limits is strict. Can not take out equality

For example,
$$\frac{1}{n} < \frac{1}{n} + \frac{1}{n^2} \qquad (n \in \mathbb{N}),$$
but
$$\lim_{n\to\infty} \frac{1}{n} = \lim_{n\to\infty} \left(\frac{1}{n} + \frac{1}{n^2}\right) = 0.$$

Definition 2.18: monotone sequences

Let $\{x_n\}$ be a sequence of real numbers.
(i) It is *increasing* if
$$x_1 \leq x_2 \leq x_3 \leq \cdots,$$
and *strictly increasing* if all these inequalities are strict.
(ii) It is *decreasing* if
$$x_1 \geq x_2 \geq x_3 \geq \cdots,$$
and *strictly decreasing* if all these inequalities are strict.
(iii) It is *monotone* if it is either increasing or decreasing.
Some authors use *nondecreasing* and *nonincreasing* for the two nonstrict notions.

If $\{x_n\}$ is increasing and $x_n \to a$, we write
$$x_n \nearrow a.$$
Similarly, $x_n \searrow a$ means that $\{x_n\}$ is decreasing and converges to a. Notice that
$$x_n \nearrow a \iff -x_n \searrow -a.$$

Theorem 2.19: monotone convergence theorem

If $\{x_n\}$ is increasing and bounded above, or decreasing and bounded below, then $\{x_n\}$ converges to a finite limit.

Proof: $E := \{x_n : n \in \mathbb{N}\}$, since E is nonempty and bounded above.

By Completeness Theorem: $\exists\ a := \sup E \in \mathbb{R}$

$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, $a - \varepsilon < x_N \leq a$

For every $n \geq N$. $a - \varepsilon < x_N \leq x_n \leq a$.

$a - \varepsilon < x_n \leq a \Rightarrow |x_n - a| < \varepsilon$. for every $n \geq N$

Hence. $x_n \to a$.

e.x. prove: if $|a| < 1$. then $a^n \to 0$

proof: let $u_n := |a|^n$ $\quad u_{n+1} = |a| \cdot u_n$

thus. u_n is decreasing. and $u_n \geq 0 \Rightarrow u_n$ is bounded below

By thm 2.19. $u_n \to L$ for some $L \geq 0$.

$u_{n+1} = |a| u_n \Rightarrow u_{n+1} \to L$.

$\lim_{n\to\infty} u_n = \lim_{n\to\infty} u_{n+1} \Rightarrow L = |a| L$.

Since $|a| \geq 0$. $L = 0$

$|a^n| \to 0 \Rightarrow a^n \to 0$

e.x. $a > 0 \Rightarrow a^{1/n} \to 1$.

(1) if $a = 1$. $a^{1/n} = 1$ for all n.

(2) if $a > 1$. $u_n := a^{1/n}$

Since $\frac{1}{n} > \frac{1}{n+1}$. u_n is decreasing.

Since $u_n > 1$ for all n. u_n is bounded below.

$\Rightarrow u_n \to L$ for some $L \geq 1$

$u_{2n} \to L$ as well $\Rightarrow L^2 = L \Rightarrow L = 1$.

(3) if $0 < a < 1$. $a^{1/n} = \frac{1}{(1/a)^{1/n}} \to 1$ by (2)

Definition 2.22: nested sets

A sequence of sets $\{I_n\}_{n\in\mathbb{N}}$ is *nested* if
$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots.$$

Theorem 2.23: nested interval property

Let $\{I_n\}$ be a nested sequence of nonempty, closed, bounded intervals. Then
$$E := \bigcap_{n=1}^{\infty} I_n$$
is nonempty. If, in addition, the lengths satisfy $|I_n| \to 0$, then E consists of a single point.

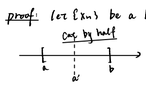
Proof. Write
$$I_n = [a_n, b_n].$$
The nesting condition implies
$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n.$$
$a_1\ a_2 \cdots a_n \to\ \ b_1\ \ b_2 \cdots b_n$

Thus $\{a_n\}$ is increasing and bounded above by b_1, while $\{b_n\}$ is decreasing and bounded below by a_1. By Theorem 2.19, there are $a, b \in \mathbb{R}$ such that
$$a_n \longrightarrow a, \qquad b_n \longrightarrow b.$$
Fix n. For every $m \geq n$,
$$a_n \leq a_m \leq b_m \leq b_n.$$
Letting $m \to \infty$ gives
$$a_n \leq a \leq b \leq b_n.$$
Consequently $[a, b] \subseteq I_n$ for every n, and hence
$$[a, b] \subseteq \bigcap_{n=1}^{\infty} I_n.$$
In particular, the intersection is nonempty.
If $|I_n| = b_n - a_n \to 0$, then
$$b - a = \lim_{n\to\infty} (b_n - a_n) = 0,$$
so $a = b$. If $x \in E$, then $a_n \leq x \leq b_n$ for every n. Taking limits gives $a \leq x \leq b = a$, and therefore $x = a$. Thus $E = \{a\}$. □

Bolzano–Weierstrass theorem

Every bounded sequence of real number has a convergent subsequence

proof: Let $\{x_n\}$ be a bounded sequence, say $\{a_n\} \subseteq [a, b]$



Cut $[a, b]$ by half to get $[a, a']$, $[a', b]$

one of those two closed interval contains ∞ many x_n.

Say that interval is $I_1 = [a_1, b_1]$

then $|I_1| = \frac{b-a}{2}$

$I_1 \subseteq I_0 = [a, b]$

Cut I_1 by half

one of those two closed interval contains ∞ many x_n.

say that interval I_2 $\quad |I_2| = \frac{b-a}{4}$.

$I_2 \subseteq I_1 \subseteq I_0$

continue in this fashion we get a nested sequence of closed interval

$$\cdots I_m \subseteq I_{m-1} \subseteq \cdots \subseteq I_1 \subseteq I_0$$

$I_m = [a_m, b_m]$ $\quad |I_m| = \frac{1}{2^m}(b-a) \longrightarrow 0$.

each I_m contains ∞ many x_n

$\Rightarrow \bigcap_{m=1}^{\infty} I_m = \{a\}$.

Now take one element $x_{n_k} \in I_k$ $\forall\ k \in \mathbb{N}$

$0 \leq |x_{n_k} - x| \leq |I_k| = \frac{b-a}{2^k} \to 0$.

The Sequeeze theorem gives $x_{n_k} \to x$

Cauchy Sequence

$\{x_n\}$ said to be Cauchy if $\forall \varepsilon > 0$ $\exists N \in \mathbb{N}$. $m, n \geq N \Rightarrow |x_m - x_n| < \varepsilon$.

remark: if $\{x_n\}$ is convergent then it is Cauchy

Theorem: $\{x_n\}$ is Cauchy $\Leftrightarrow$ $\{x_n\}$ convergents to some point $\alpha \in \mathbb{R}$

Suppose $\{x_n\}$ is Cauchy (2) "B-W Theorem"

(1) x_n is bounded $\Rightarrow \exists\ \{x_{n_k}\}$ is converge to a.

$\Rightarrow x_n$ is converge to a since $\{x_n\}$ is Cauchy

For $\varepsilon = 1$ $\exists N > 0$ $\quad m, n \geq N \Rightarrow |x_m - x_n| < 1$.

$\Rightarrow |x_m - x_N| > 1$ for any $m \geq N$

$|x_m| = |x_m - x_N + x_N| \leq |x_m - x_N| + |x_N| < 1 + |x_N|$

So $|x_m| \leq \max\{|x_1|, |x_2|, \cdots, |x_{N-1}|, 1 + |x_N|\}$ for any m

$\Rightarrow \{x_n\}$ is bounded

(2) $\forall \varepsilon > 0$ $\exists k_0$. $\quad k \geq k_0 \Rightarrow |x_{n_k} - a| < \frac{\varepsilon}{2}$.

Since x_n is Cauchy. $\exists N > 0$ $\quad m, n \geq N \Rightarrow |x_m - x_n| < \frac{\varepsilon}{2}$

Take $k_1 \geq k_0$ such that $n_{k_1} \geq N$.

Then $\forall m \geq N$ we have

$|x_m - a| \leq |x_m - x_{n_{k_1}}| + |x_{n_{k_1}} - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

$\{x_n\}$, $|x_n - x_{n+1}| \leq \frac{1}{2^n} \Rightarrow x_n$ is Convergent

$|x_n - x_m| = |x_n - x_{n+1} + x_{n+1} - x_{n+2} \cdots + x_{m-1} - x_m|$

$\leq |x_n - x_{n+1}| + |x_{n+1} - x_{n+2}| + \cdots + |x_{m-1} - x_m|$

$\leq \frac{1}{2^n} + \frac{1}{2^{n+1}} + \cdots + \frac{1}{2^{m-1}}$

$\leq \frac{1}{2^n}$

remark: $\{x_n\}$ satisfying $x_{n+1} - x_n \to 0$ need not to be cauchy

$x_n = \log n$

$x_{n+1} - x_n = \log(n+1) - \log n = \log \frac{n+1}{n} \to \log 1 = 0$ as $n \to \infty$

However, since $\log n$ doesn't converge. It is not Cauchy

Limit Superior & inferior

$\{x_n\}_{n=1}^{\infty}$

$S_n = \sup_{k \geq n} x_k = \sup\{x_k : k \geq n\}$

$t_n = \inf_{k \geq n} x_k = \inf\{x_k : k \geq n\}$

e.x. $\{\frac{1}{n}\}_{n=1}^{\infty}$ $\quad S_n = \sup_{k \geq n} x_k = \sup_{k \geq n} \frac{1}{k} = \frac{1}{k}$.

$t_n = \inf_{k \geq n} x_k = \inf_{k \geq n} \frac{1}{k} = 0$

remark: if $A \subseteq B$, then:

$\Rightarrow \sup A \leq \sup B$. $\inf A \geq \inf B$

since $\{y_k : k \geq n+1\} \subseteq \{x_k : k \geq n\}$

$\Rightarrow S_{n+1} \leq S_n$. $t_{n+1} \geq t_n$

$S_n \geq S_{n+1} \geq S_{n+2} \cdots$

$t_n \leq t_{n+1} \leq t_{n+2} \cdots$



$\Rightarrow \lim_{n\to\infty} S_n = a$. $\quad (a \in \mathbb{R} \cup \{+\infty\})$

$\lim_{n\to\infty} t_n = b$ $\quad (b \in \mathbb{R} \cup \{-\infty\})$

Limit supremum / infimum

$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} (\sup_{k \geq n} x_k)$

$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} (\inf_{k \geq n} x_k)$

e.x. for $x_n = (-1)^n$ $\quad \sup_{k \geq n} (-1)^k = \sup\{-1, 1\} = 1$.

$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} (\sup_{k \geq n} (-1)^k) = 1$.

$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} (\inf_{k \geq n} (-1)^k) = -1$

$\inf\{-1, 1\} = -1$

e.x. $x_n = 1 + 1/n$.

$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} (\sup_{k \geq n} (1 + \frac{1}{k})) = \lim_{n\to\infty} (1 + \frac{1}{n}) = 1$.

$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} (\inf_{k \geq n} (1 + 1/n)) = \lim_{n\to\infty} 1 = 1$.

Thm Suppose $\{x_n\}$ is a bounded sequence

$\limsup_{n\to\infty} x_n = a \in \mathbb{R}$ $\quad \liminf_{n\to\infty} x_n = b \in \mathbb{R}$

(1) if $r_1 \leq x_n \leq r_2$ $\forall\ n \in \mathbb{N}$ then $b \leq r_1 \leq r_2 \leq a$ proof: $r_1 \leq x_n \leq r_2$ $\forall\ n \in \mathbb{N}$.

(2) $\exists$ subsequences. $\{x_{n_k}\}, \{x_{n_{k'}}\}$ such that: $a = \lim_{n\to\infty} \sup x_n = \lim_{n\to\infty} (\sup_{k \geq n} x_k) \leq r_2$; $b = \lim_{n\to\infty} \inf x_n = \lim_{n\to\infty} (\inf_{k \geq n} x_k) \geq r_1$

$\lim_{k\to\infty} x_{n_k} = a$ $\quad \lim_{k\to\infty} x_{n_{k'}} = b$

proof: since $\lim_{n\to\infty} \sup x_n = a$.

$S_n = \sup\{x_k : k \geq n\}$, $S_n \searrow a$.

$\{n_k\}$. bounded

$x_{n_1} \leq S_{n_0}$

For $n_0 = 0$ $\quad S_1 = S_{n_0+1} = \sup\{x_k : k \geq 1\}$.

$\exists\ n_1 \in \mathbb{N}$ $\quad S_{n_0+1} - 1 < x_{n_1} \leq S_{n_0+1}$.

$\exists\ n_2 \geq n_1 + 1$ $\quad S_{n_1+1} - \frac{1}{2} < x_{n_2} < S_{n_1+1}$

$\exists\ n_3 \geq n_2 + 1$ $\quad S_{n_2+1} - \frac{1}{3} < x_{n_3} < S_{n_2+1}$

$\cdots$

$S_{n_{k-1}+1} - \frac{1}{k} < x_{n_k} \leq S_{n_{k-1}+1}$, $\forall\ k \in \mathbb{N}$

$k \to \infty$. $\lim_{n\to\infty} S_n = a \Rightarrow S_{n_k+1} \to a$ as $k \to \infty$.

Sequeeze thm: $\lim_{k\to\infty} x_{n_k} = a$

(3) if $\{x_{n_k}\}$ is a convergent subsequence of $\{x_n\}$.

Then $b \leq \lim_{k\to\infty} x_{n_k} \leq a$

proof: Suppose $\lim_{k\to\infty} x_{n_k} = x$. $\quad N \to n_k$, $k \to \infty$. $x_{n_k} \to x$

$\forall\ N \in \mathbb{N} \cdot \exists\ k_N > 0$. $\quad k \geq k_N \Rightarrow n_k \geq N$.

$\Rightarrow \inf\{x_n : n \geq N\} \leq x_{n_k} \leq \sup\{x_n : n \geq N\}$ $\forall\ k$

$\Rightarrow \inf\{x_n : n \geq N\} \leq x \leq \sup\{x_n : n \geq N\}$

$k \to \infty$. $N \to \infty$.

$b \leq x \leq a$